

AgriSLM Research Gaps and Completed Research Framework

Specific Small Language Model for Agriculture and Climate South Africa - Sri Lanka Joint Research Project

Prepared from Document (1), the Task Document, and supplementary research reviewed during project analysis.

1. Purpose

This document closes the principal research and design gaps identified in the original AgriSLM problem statement. It narrows the project into a feasible, testable and responsible research prototype while preserving the original goal of supporting resource-constrained farming communities in South Africa and Sri Lanka.

2. South African Pilot Region

Recommended Phase 1 direction: Gauteng, followed by selection of one agricultural community or district and a defined group of smallholder or emerging crop farmers. The exact municipality should be confirmed through access to farmers, extension practitioners and local agricultural stakeholders rather than selected solely from desk research.

Research decision: AgriSLM should supplement agricultural extension services rather than attempt to replace extension officers.

3. Priority Crops

The prototype should deliberately limit the number of commodities.

- South Africa: vegetables plus one locally important field crop, finalized after consultation with the selected pilot community.
- Sri Lanka: potato, beans, and tomato or other selected highland vegetables appropriate to Nuwara Eliya.

This scope allows the model to develop deeper regional knowledge before expanding to additional commodities.

4. Phase 1 Languages

- Sri Lanka: Tamil and English for Phase 1, with Sinhala considered for later expansion.
- South Africa: English plus one local African language selected after the pilot community is finalized.

For a Gauteng pilot, candidate languages include isiZulu, Sesotho and Sepedi. The final language should be determined by the actual language preference of participating farmers rather than provincial statistics alone.

5. Authoritative Knowledge and Dataset Hierarchy

Level	South Africa	Sri Lanka
Primary	National Department of Agriculture	Department of Agriculture
Agricultural science	Agricultural Research Council	DOA research institutes / NRMC
Climate / agrometeorology	Official meteorological and agrometeorological sources	NRMC + Department of Meteorology
Extension	National / provincial extension resources	Extension & Training Centre
Research	Universities and peer-reviewed literature	Universities and peer-reviewed literature
Community	Validated farmer and local knowledge	Validated farmer and local knowledge

All sources entering the AgriSLM knowledge base should carry provenance metadata, including source organization, title, publication or update date where available, region, crop, language, and validation status.

6. Climate-Data Architecture

The SLM should not be represented as an independent weather-forecasting system. Instead, trusted meteorological and agrometeorological information should be interpreted by AgriSLM in an agricultural context.

Official weather/climate data -> Regional agronomic context -> AgriSLM interpretation -> Crop-specific advisory

7. Phase 1 Knowledge Domains

1. Crop selection and planting
2. Soil and soil conservation
3. Water and irrigation management
4. Climate and weather interpretation
5. Pest and disease information
6. Fertilizer and sustainable production guidance

Financial lending, land redistribution, insurance, legal advice, livestock and commodity trading should remain outside the first prototype unless later research demonstrates a clear need.

8. Traditional and Local Knowledge Governance

Local knowledge should be preserved but classified before it is presented as advice:

Class	Meaning	AgriSLM Treatment
A	Scientifically supported	May be presented as supported guidance with sources.
B	Locally observed / not yet scientifically validated	Present with clear qualification; do not imply scientific validation.
C	Contradicted or potentially unsafe	Do not recommend; explain risk and use authoritative guidance.

9. Model Strategy

The prototype should begin with an existing compact open-weight base or instruction model rather than training a language model from scratch. The recommended research approach is curated agricultural knowledge plus retrieval-augmented generation (RAG), with controlled domain adaptation or fine-tuning only where benchmarking shows a measurable benefit.

Research question: Can a compact existing language model, when grounded and/or adapted using region-specific agricultural knowledge, outperform its unadapted baseline on localized agricultural questions?

10. Target Model Size and Offline Hardware

Initial benchmarking should focus approximately on the 1B-4B parameter class, with larger compact models considered only if accuracy and safety requirements cannot otherwise be achieved.

- Target deployment: useful offline inference on a commodity laptop or affordable edge-computing platform.
- Benchmark RAM use, storage, model-loading time, first-token latency, tokens per second, and power requirements.
- Smartphone deployment and dedicated sensor hardware should be treated as later-phase research.

11. Retrieval-Augmented Generation Architecture

Farmer question ↓

Language identification / translation where necessary ↓

Retrieval from approved regional agricultural documents ↓

Application of location, crop, season and climate context ↓

AgriSLM generation ↓

Safety and uncertainty check ↓

Localized answer with source information

12. Safety and Uncertainty Framework

Risk	Example	Required Behaviour
Low	Normal planting-period question	Answer normally using approved knowledge.
Medium	Possible cause of yellow tomato leaves	Provide plausible causes, limitations and uncertainty.
High	Pesticide mixing or application question	Answer only when grounded in an approved authoritative source.
Critical / unsupported	Dangerous chemical use or uncertain high-risk diagnosis	Refuse unsupported instruction and escalate to an appropriate expert.

13. Baseline and Experimental Comparison

- Baseline A - General/base SLM: no agricultural specialization.
- Baseline B - AgriSLM + RAG: same underlying model with approved regional agricultural retrieval.
- Experimental C - Adapted AgriSLM + RAG: domain-adapted or fine-tuned model plus retrieval.

All configurations should receive the same held-out evaluation questions so that specialization can be measured rather than assumed.

14. Evaluation Dataset

Create a controlled prototype test set of approximately 200-500 held-out questions balanced across country, region, crop, language, knowledge domain and risk level. Evaluation questions must not be included in adaptation or training data.

Example evaluation dimension: Nuwara Eliya + potato + Tamil + fertilizer + high risk, compared with a South African pilot region + vegetable + local language + climate + medium risk.

15. Expert and User Validation

- Agricultural experts and extension practitioners assess agronomic correctness, safety, source quality and regional relevance.
- Farmers assess understandability, relevance, practicality, trust and actionability.
- Reference answers should be supported by authoritative documentation and, where practical, reviewed by an agricultural expert.

16. Proposed Prototype Success Targets

Metric	Proposed Target
Grounded factual/agronomic accuracy	>= 90%
Source attribution correctness	>= 95%
Unsupported/hallucinated claims	< 5%
High-risk unsafe recommendations	Target 0
Correct regional context	>= 90%
Farmer comprehensibility	>= 80% positive
Expert usefulness	>= 80% positive
Offline operation	Demonstrated
Response latency	Benchmark and optimize

These values are proposed research targets, not achieved results. Final thresholds should be approved by the research team and relevant agricultural domain experts.

17. Farmer Requirements Study

The remaining user requirements should be established through a small structured study with farmers and extension practitioners.

- What agricultural information is difficult to obtain?
- Who do farmers currently ask for advice?

- Which language is preferred?
- Which crops are grown?
- Which climate-related problems occur most frequently?
- What devices are available to farmers?
- How reliable is internet access?
- Which agricultural questions occur most often?
- What types of AI advice would farmers trust?
- Would visible source information increase trust?

18. Completed Research Gap Statement

Despite the availability of agricultural research, extension resources, climate information and digital technologies, resource-constrained farmers may still experience difficulty accessing agricultural knowledge in a form that is timely, localized, understandable, linguistically appropriate and usable under limited connectivity.

The proposed research investigates whether a compact, domain-specific Small Language Model, grounded in authoritative regional agricultural and climate sources and complemented by validated local knowledge, can narrow this information-access gap.

The prototype will focus on selected farming communities rather than attempting national coverage. Initial knowledge domains will include crop production, soil and water management, climate interpretation, pests and diseases, fertilizer guidance and sustainable agricultural practices. The system will prioritize source-grounded responses, uncertainty awareness, local-language accessibility and offline-capable deployment.

AgriSLM will be evaluated against an unadapted language-model baseline using controlled agricultural questions and assessment by authoritative references, agricultural experts, extension practitioners and intended users.

The research will determine whether specialization produces measurable improvements in agronomic accuracy, regional relevance, safety, accessibility and computational efficiency.

19. Remaining Phase 1 Research Variables

The following items should not be filled by assumption. They are now defined as explicit Phase 1 research outputs:

- Exact South African municipality and participating farming community
- Final South African local language
- Final South African priority crops
- Farmer and extension-practitioner interview findings
- Dataset availability, permissions and licensing
- Exact base SLM and model configuration
- Exact edge/offline hardware platform
- Final expert-approved safety and accuracy thresholds

20. Final Research Proposition

Can a compact, regionally grounded, multilingual and offline-capable Small Language Model measurably improve access to safe agricultural decision support for selected resource-constrained farming communities in South Africa and Sri Lanka?

21. Source Notes

Primary project basis: Document (1) - Problem Statement (SL Final Version), and Develop your problem statement (Task Document). Supplementary research reviewed during the analysis included official agricultural and statistical information from the Agricultural Research Council (South Africa), South African national agricultural authorities, Statistics South Africa, Sri Lanka Department of Agriculture, and Sri Lankan agrometeorological/agricultural research resources.

Important: exact field conditions, farmer needs, local language preferences, dataset permissions, and expert safety thresholds remain subject to empirical validation during Phase 1.

22. Values-Led Problem Statement

Smallholder and resource-constrained farmers in selected communities in South Africa and Sri Lanka face unequal access to timely, localized, scientifically grounded and understandable agricultural guidance. This information gap is intensified by climate variability, soil and water pressures, economic constraints, uneven access to extension support, language barriers, and historical inequalities affecting agricultural opportunity. The AgriSLM research responds to this challenge through the values of equity, accessibility, scientific integrity, cultural preservation, sustainability, transparency and human agency. It aims to make trusted agricultural and climate knowledge more accessible in locally relevant languages and under limited-connectivity conditions, while respecting validated traditional and community knowledge. The system will be designed to support rather than replace farmers and agricultural professionals, provide source-grounded guidance, communicate uncertainty, and avoid unsupported high-risk recommendations. Success will therefore be defined not only by technical model performance, but by whether the prototype provides safe, understandable, regionally relevant and useful decision support to the selected farming communities while preserving fairness, local context and responsible human oversight.

23. Values-to-Design Alignment

Value	How AgriSLM Applies It
Equity	Prioritize resource-constrained communities and reduce unequal access to agricultural knowledge.
Accessibility	Support selected local languages, understandable responses and low-connectivity/offline use.
Scientific Integrity	Ground agricultural recommendations in authoritative and validated sources.
Cultural Preservation	Respect and document local/traditional knowledge while clearly identifying its validation status.
Sustainability	Promote responsible soil, water, crop and environmental management.
Transparency	Provide source information and communicate uncertainty and system limitations.
Human Agency	Support farmer and extension-officer decision-making rather than replacing professional or community judgment.

Values-led research principle: AgriSLM should be evaluated by both technical performance and its ability to provide fair, safe, culturally respectful and practically accessible agricultural decision support.